231/3

NAME……………………………………………………………ADM.NO………………………

SIGNATURE…………………………………………… DATE………………………………

ROYAL

MARCH / APRIL - 2025

BIOLOGY

PAPER 3 (PRACTICAL)

TIME: 1 ¾ HOURS

INSTRUCTIONS TO CANDIDATES

Answer all the questions in the spaces provided.

You are required to spend the first 15 minutes of the 1 ¾ hours allowed for this paper reading the whole paper carefully before commencing your work.

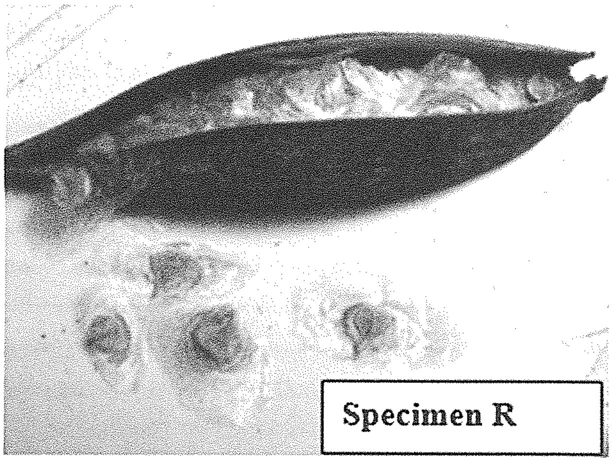
Additional pages must not be inserted.

FOR EXAMINERS USE ONLY.

Question	Maximum score	Candidates score
1	15	
2	15	
3	10	
Total score	**40**	

This paper consists of 5 printed pages. Candidates should check to ensure that all pages are printed as indicated and no questions are missing

Q1. You are provided with **specimen Q** and Photograph of **specimen R**.



Specimen R

ai) Which plant organ are specimen **Q** and **R**? (1mk)

....

ii) Give the classification name for Specimen **Q**: - (1mks)

....

b) With a reason, State the mode of seed dispersal for specimens **Q** and **R** (4mks)

i) Specimen **Q** ....

Reason

....

....

ii) Specimen **R** ....

Reason

....

....

c) Make a cross section of specimen **Q** to obtain two halves.

i) Draw and label the cut section of one half. (4mks)

ii)State the placentation of specimen **Q** ………………………………………………… (1mk)

c) You are given that 5 drops of 0.1 % solution of Ascorbic Acid are required to decolorize $2cm^3$ of 1% DCPIP.

Squeeze some juice from specimen Q into a 100ml beaker.

(i)Using a measuring cylinder put $2cm^3$ of DCPIP solution into a test-tube.

Add the juice dropwise and record the number of drops used to decolorize DCPIP in the table below. (1mks)

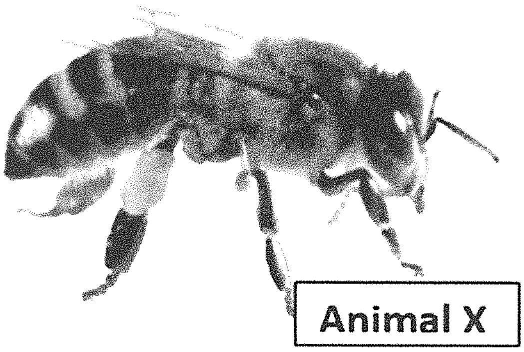
JUICE /LIQUID	No. of drops to decolorize DCPIP
0.1% ascorbic acid	**5**
juice of specimen K	

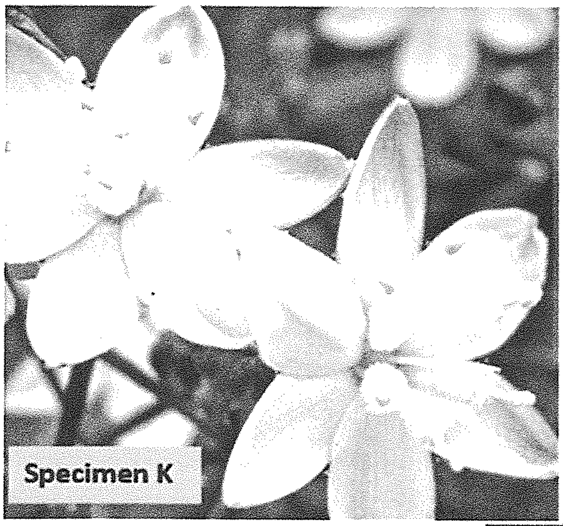
ii) Calculate the percentage of ascorbic acid in the juice of specimen Q (2mks)

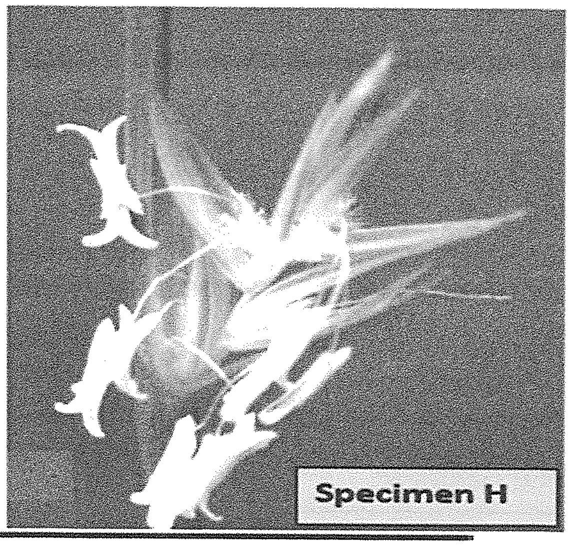
d) What disease is caused by the deficiency of Ascorbic acid in man (1mk)

....

Q2. Below are diagrams of two flowers and an animal.







ai) What is the role of the **Animal X** in reproduction in plants? (1mk)

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ii) State two observable features that adapt animal **X** to the role you have stated in a(i) above (2mks)

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bi) Giving a reason, name the sub-division to which plants from which both specimen **H** and **K** were obtained. (2mks)

Sub-division ---

Reason: -

----

ii) To which class does specimen **K** belong? (1mk)

----

iii) Give a reason for your answer in (ii) above. (1mk)

Ci) Write in the spaces below the term anemorphilous or entomorphilous to best describes flower of specimen **H** and **K.** (2mks)

Specimen H ....

Specimen K ....

ii) Compare the adaptations of the flowers from the two specimens to pollination. (3mks)

Specimen H	**Specimen K**

iii) State **three** visible features of leaves of specimen H that can be used to classify it into its class. (3mks)

Q3. You are provided with **Solution A**, distilled water and a piece of visking tubing.

Rub the visking tubing under water to open it then, using a piece of thread, tightly tie one end of the tubing.

Half fill the visking tubing with solution A and tightly tie the open end with a piece of thread ensuring there is no leakage.
Immerse the tubing in a beaker half filled with distilled water.

Leave the set up for 30minutes.

ai) Which physiological process is being investigated? (1mk)

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ii) Why was the visking tube used in this experiment? (1mks)

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bi) Remove the visking tube from the beaker and record your observation. (1mks)

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ii) Account for the observation recorded above. (3mks)

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ci) Briefly describe what would happen to the red blood cell if placed in solution A. (2mks)

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ii) Explain why Onion epidermal cell will not lose its shape when placed in solution A. (2mks)

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BIOLOGY PAPER 3 REQUIREMENTS

Q1 – **Specimen Q** - 1 small ripe Orange.

- 1 test-tubes
- 1 dropper
- Scalpel.
- White tile
- 1% DCPIP
- 100ml beaker
- Measuring cylinder
-

Q3 - 10ml **Solution A** – Concentrated Salt solution.

- Distilled water
- 100ml beaker
- 1 visking tube
- 2 pieces of thread.